# HYBRID SOLAR POWERED RC PLANE

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| **PROJECT NAME:** | Hybrid Solar Powered RC Plane |
| **PROJECT GROUP:** | Kunal Pisudde |

## WHY THIS PROJECT ?

- Conventional RC aircraft have limited flight endurance
- Inefficient power utilization causes rapid battery discharge
- Direct battery supply leads to voltage mismatch and energy losses
- Need for energy-efficient and sustainable UAV systems
- Increasing demand for long-duration aerial missions such as monitoring and surveillance

## INTRODUCTION

- This project focuses on extending the flight time of an RC aircraft
- Implements an efficient energy management system
- Integrates solar cells, Li-Po battery, voltage regulation and power control
- Ensures stable and reliable power delivery under varying flight conditions
- Improves overall efficiency, endurance and battery life

## KEY FEATURES

- Hybrid solar-assisted power system
- High-efficiency buck-boost DC-DC voltage regulator
- Battery protection against:
  1. Overcharging
  2. Over-discharging
  3. Excess current
- Lightweight and compact system
- Improved flight endurance and reliability

## ARCHITECTURE

- Solar panels convert sunlight into electrical energy
- Voltage Regulator Module:
  - Steps up/down voltage automatically
  - Maintains constant output
- Li-Po battery stores energy and provides backup power
- ESC regulates motor speed efficiently
- BLDC motor converts electrical energy into thrust
- Ensures safe and efficient energy utilization

## RESULTS

- Significant increase in flight duration
- Stable voltage supply during changing load conditions
- Reduced battery stress and discharge rate
- Improved system efficiency compared to conventional RC aircraft
- Reliable and smooth flight performance achieved

## FUTURE SCOPE

1. **Use of advanced and lightweight solar panels**
2. **Implementation of smart energy management algorithms**
3. **Development of fully sustainable aerial platforms**